

ZONE 1 BATTERY / PROTECTION

B1 / BMS / F1 [S2]

B1 + to BMS inputN-BAT+

BMS output to F1 INN-BMS+

F1 OUT -> F2 IN / F3 IN / K1 MAIN INN-PROTECTED12+

B1 / BMS negative -> U3 T4N-TRACTION-

F1, F2 and F3 ratings: TBD / VERIFY.

W-PWR-001 to 008 define each branch on S2.

BMS terminal names are functional only; H03 applies.

No added connection to N-LOGIC-GND.

The shared protected distribution has ONE canonical net name.

ZONE 2 LOGIC POWER CONVERSION

PS1 FOUR DISTINCT DC-DC TERMINALS [S2]

IN+ <- F2 OUTN-LOGIC12+

IN- source unassigned / HOLDPS1-IN-RETURN-HOLD

OUT+ -> U1 / U2 / LD2450N-LOGIC5+

OUT- -> approved logic referencesN-LOGIC-GND

VERIFY EXACT CONVERTER (H05).

Do not assume IN- and OUT- are common.

No IN-/OUT- bridge is drawn or present in the database.

N-LOGIC12+ exists only AFTER F2.

W-LOGIC-001 to 004; W-GND-001 to 009 on S2.

ZONE 5 SENSORS / GNSS / CAMERA

SENSOR CONNECTORS [S3]

LD2450 Pin 1 VCCN-LOGIC5+

LD2450 Pin 2 GNDN-LOGIC-GND

LD2450 Pin 3 TX -> U1 Pin 10RADAR-TX

LD2450 Pin 4 RX <- U1 Pin 8RADAR-RX

MLX90640 / BNO055 SDAI2C-SDA

MLX90640 / BNO055 SCLI2C-SCL

MLX90640 / BNO055 VCCHOLD - UNASSIGNED

GNSS1 USB serial -> U1 USBGNSS-USB

CAM1 USB/UVC -> U1 USBCAMERA-USB

Sensor GND -> N-LOGIC-GND: HOLD / PHYSICAL PIN VERIFY. VCC, pullups and addresses: HOLD.

UART: 256000 baud, 8 data bits, no parity, 1 stop bit.

GNSS serial path/baud/NMEA: verify during commissioning.

ZONE 8 E-STOP / K1 ISOLATION

F3 / ES1 / CD1 / K1 [S2]

F3 OUT -> ES1 NC INN-ESTOP-FUSED+

ES1 NC OUT -> CD1 INN-ESTOP-SWITCHED+

CD1 OUT -> K1 COIL AK1-COIL+

K1 COIL B / suppression returnK1-COIL-

K1 MAIN OUT -> U3 T3N-TRACTION12+

SC1 OUT -> U2 J1-19 GPIO13ESTOP-AUX-STATUS

Coil suppression is across the TWO coil terminals.

Coil return and CD1 circuit remain HOLD (H11/H12/H14).

K1 auxiliary conditioning is diagnostic only (H15).

Independent physical E-stop; no GPIO safety-current path.

ZONE 3 RASPBERRY PI 4

U1 CONTROLLED HEADER PINS [ALL 40: S3]

Pin 1 3V3NO EXTERNAL POWER

Pin 2 5VN-LOGIC5+

Pin 3 GPIO2 SDAI2C-SDA

Pin 5 GPIO3 SCLI2C-SCL

Pin 6 GNDN-LOGIC-GND

Pin 8 GPIO14 TXRADAR-RX

Pin 10 GPIO15 RXRADAR-TX

Pin 20 GNDN-LOGIC-GND

Pi Pin 1: NO EXTERNAL POWER.

GNSS1 and CAM1 use USB assembly boundaries.

Pi/ESP32 serial is a configured transport boundary [S5].

ZONE 7 MOTORS / HALL ENCODERS

M1 / M2 POWER AND SEPARATE ENCODERS [S4]

U3 T2 -> M1 AM1A

U3 T1 -> M1 BM1B

U3 T5 -> M2 AM2A

U3 T6 -> M2 BM2B

ENC-L A -> U2 J1-4 GPIO4ENC-L-A

ENC-L B -> U2 J1-5 GPIO5ENC-L-B

ENC-R A -> U2 J1-6 GPIO6ENC-R-A

ENC-R B -> U2 J1-7 GPIO7ENC-R-B

Encoder VCC - no connectionHOLD - H07/H08

Encoder GND: HOLD / PHYSICAL PIN VERIFY; interface conditional.

No encoder connector pin numbers or wire colors invented.

Wheel response does not establish ground speed.

ZONE 6 MDD10A REV2.0

U3 LOGIC HEADER / POWER BLOCK [S4]

P1 GNDN-LOGIC-GND

P2 PWM2ESP-PWM2

P3 DIR2ESP-DIR2

P4 PWM1ESP-PWM1

P5 DIR1ESP-DIR1

T1 M1BM1B

T2 M1AM1A

T3 POWER+N-TRACTION12+

T4 POWER-N-TRACTION-

T5 M2AM2A

T6 M2BM2B

T4: HIGH CURRENT RETURN. T3: HOLD pending physical E-stop test.

ZONE 4 ESP32-S3 / ENCODERS

U2 J1 ALLOCATED POSITIONS [ALL 22 ROWS: S4]

J1-4 GPIO4ENC-L-A

J1-5 GPIO5ENC-L-B

J1-6 GPIO6ENC-R-A

J1-7 GPIO7ENC-R-B

J1-15 GPIO9ESP-PWM1

J1-16 GPIO10ESP-DIR1

J1-17 GPIO11ESP-PWM2

J1-18 GPIO12ESP-DIR2

J1-19 GPIO13ESTOP-AUX-STATUS

J1-21 5VN-LOGIC5+

J1-22 GNDN-LOGIC-GND

All other J1 positions: UNALLOCATED / DO NOT CONNECT.

ZONE 9 COMMUNICATION / HMI

SOFTWARE RELATIONSHIPS ONLY - see S5.

U1 -> FastAPI -> WebSocket -> React/TypeScript dashboard.

Dashboard: monitoring only; no motion authority.

GNSS -> normalized location -> vehicle pose -> map.

GNSS/map do not command motors or override PV-SOE.

Control: Pi -> Protocol V1 -> ESP32 supervisor.

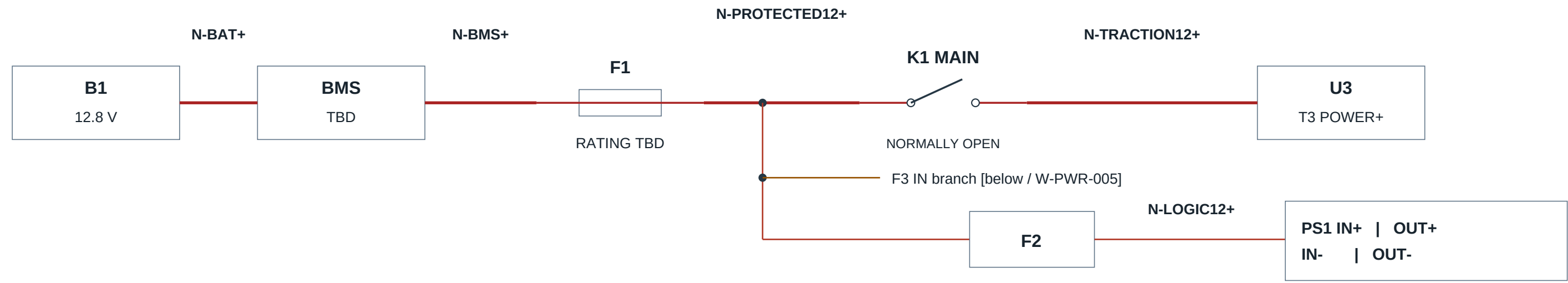
Framing: COBS / CRC-32C / canonical CBOR.

ACK/NACK/STATUS do not prove physical actuation.

Frozen software: 2d319e7; QA: 962365e.

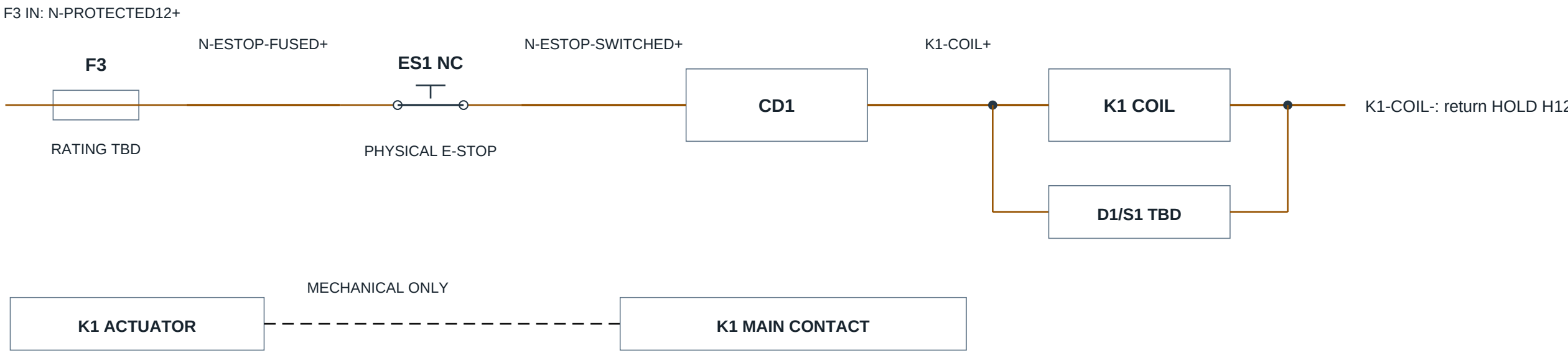
Every master net reference resolves on S2-S5.

POWER TREE - CONDUCTOR SEGMENTS AND SERIES DEVICES



Common protected supply = N-PROTECTED12+. K1 switches only its traction branch.
N-TRACTION- is the B1/BMS high-current return to U3 T4. Logic return remains separate.
PS1 IN-: PS1-IN-RETURN-HOLD -> OPEN / HOLD H05. PS1 OUT-: N-LOGIC-GND.
Wire lanes below are the complete pin-to-pin definition; this tree repeats their net relationships.

INDEPENDENT PHYSICAL E-STOP / K1 COIL / AUXILIARY



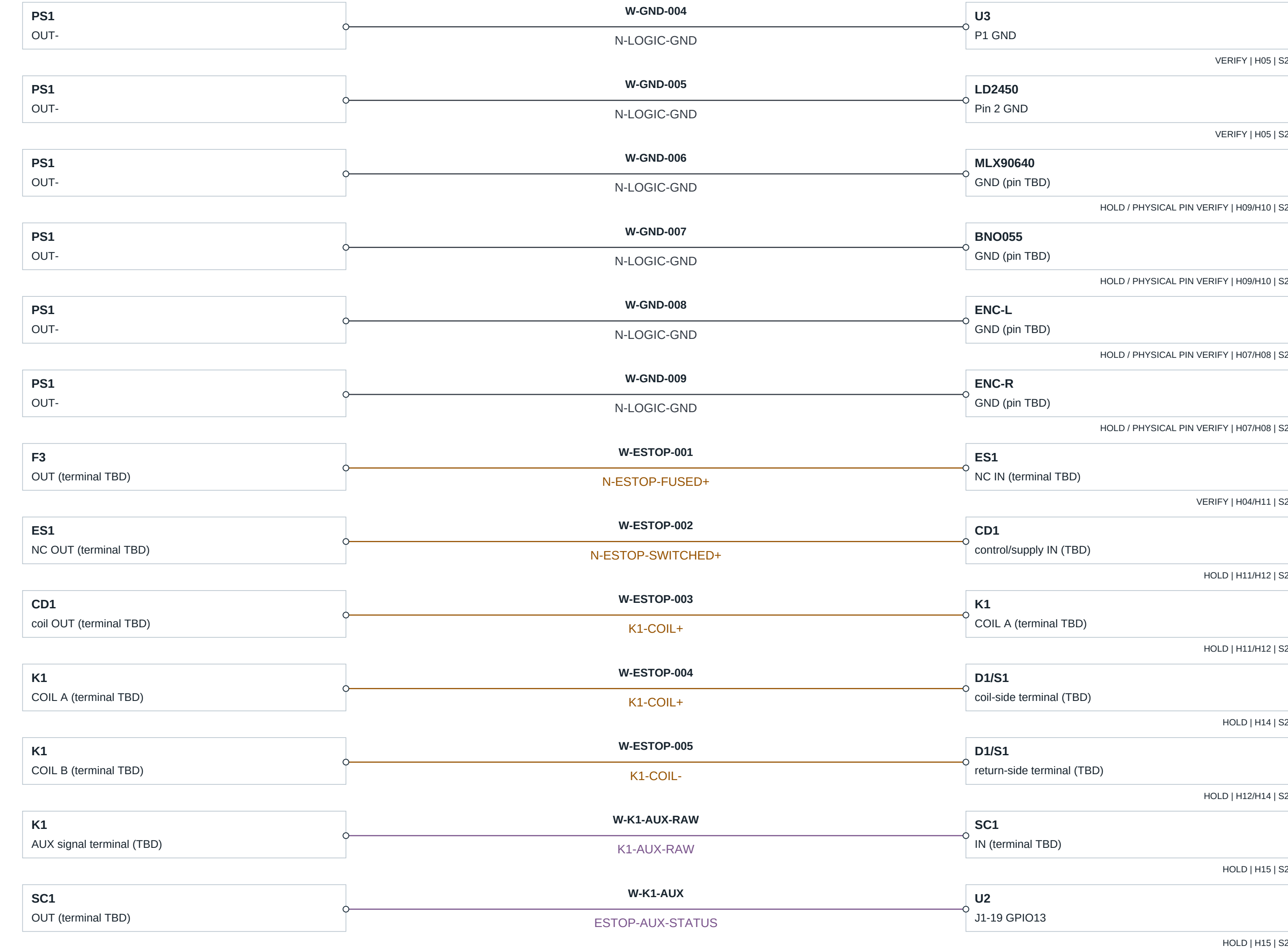
Mechanical linkage: actuator to contact only; not a conductor or coil return.

K1 COIL A / COIL B / MAIN IN / MAIN OUT ARE FUNCTIONAL INTERFACES.
PHYSICAL CONTACTOR TERMINAL NUMBERS MUST BE TAKEN FROM THE PURCHASED CONTACTOR DATASHEET AND VERIFIED BEFORE CONNECTION.

DESIGN HOLD — DO NOT ENERGIZE

Exact K1 coil voltage; exact K1 coil current; contactor driver implementation;
coil return; suppression polarity/component must be verified from the purchased contactor and driver before energization. Net + / - names do not establish coil polarity.

K1 AUX -> SC1 -> GPIO13: DIAGNOSTIC ONLY / NOT PRIMARY E-STOP; excitation HOLD.



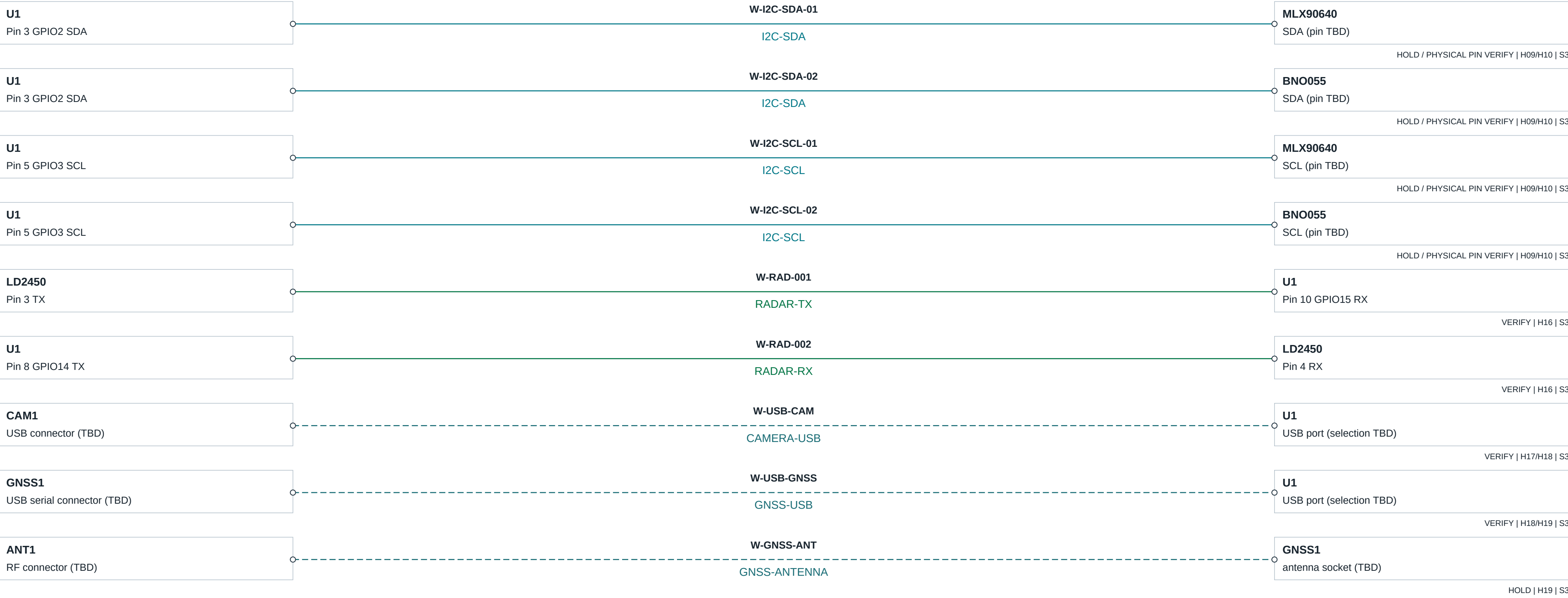
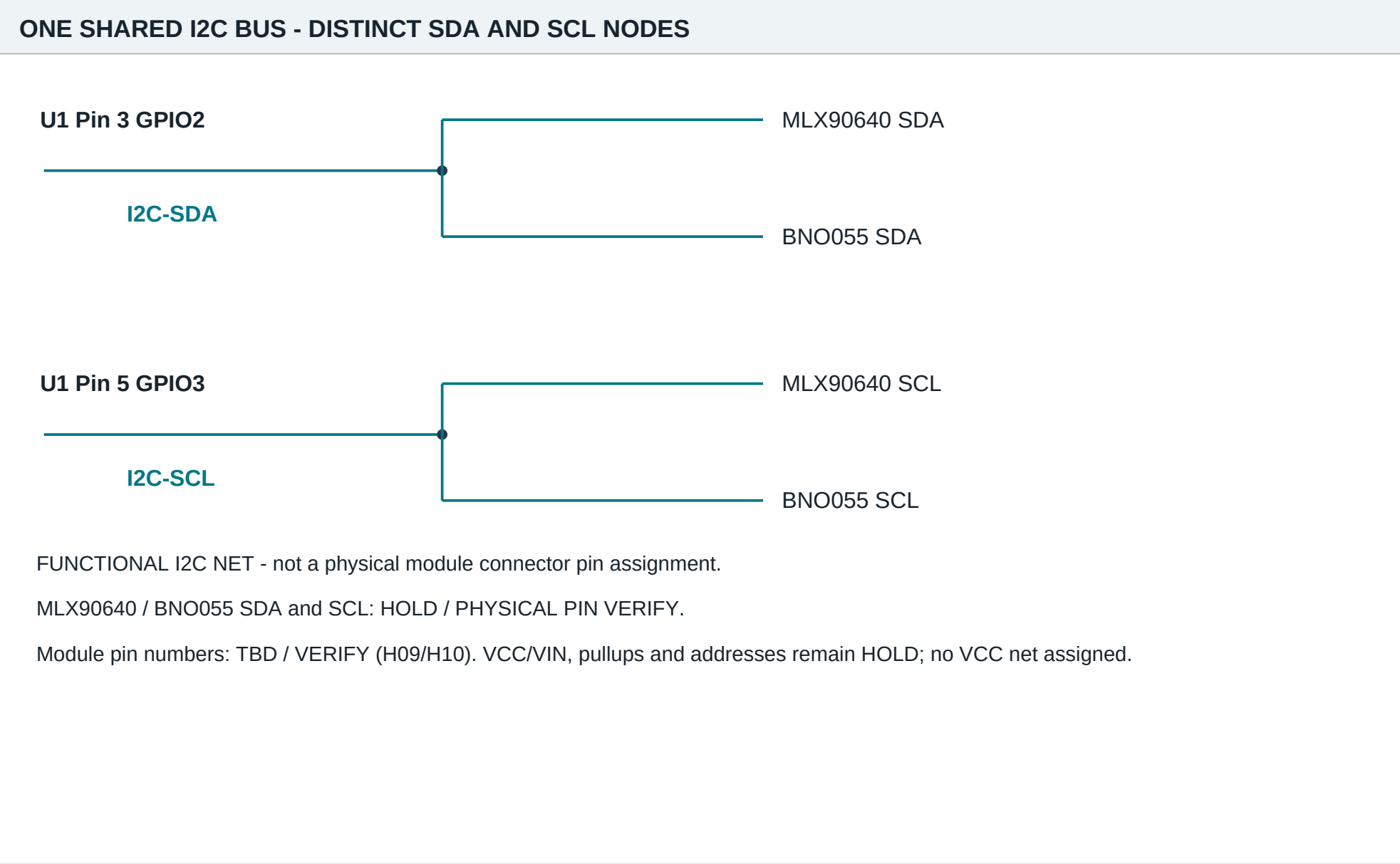
U1 RASPBERRY PI 4 - ALL 40 PHYSICAL HEADER POSITIONS			
Physical Pin	Known Function	TARK Allocation	Status
Pin 1	3V3	NO EXTERNAL POWER	DO NOT CONNECT
Pin 2	5V	N-LOGIC5+	CONTROLLED / VERIFY
Pin 3	GPIO2 SDA	I2C-SDA	CONTROLLED / VERIFY
Pin 4	5V	UNALLOCATED	DO NOT CONNECT
Pin 5	GPIO3 SCL	I2C-SCL	CONTROLLED / VERIFY
Pin 6	GND	N-LOGIC-GND	CONTROLLED / VERIFY
Pin 7	GPIO4	UNALLOCATED	DO NOT CONNECT
Pin 8	GPIO14 TX	RADAR-RX	CONTROLLED / VERIFY
Pin 9	GND	UNALLOCATED	DO NOT CONNECT
Pin 10	GPIO15 RX	RADAR-TX	CONTROLLED / VERIFY
Pin 11	GPIO17	UNALLOCATED	DO NOT CONNECT
Pin 12	GPIO18	UNALLOCATED	DO NOT CONNECT
Pin 13	GPIO27	UNALLOCATED	DO NOT CONNECT
Pin 14	GND	UNALLOCATED	DO NOT CONNECT
Pin 15	GPIO22	UNALLOCATED	DO NOT CONNECT
Pin 16	GPIO23	UNALLOCATED	DO NOT CONNECT
Pin 17	3V3	UNALLOCATED	DO NOT CONNECT
Pin 18	GPIO24	UNALLOCATED	DO NOT CONNECT
Pin 19	GPIO10	UNALLOCATED	DO NOT CONNECT
Pin 20	GND	N-LOGIC-GND	CONTROLLED / VERIFY
Pin 21	GPIO9	UNALLOCATED	DO NOT CONNECT
Pin 22	GPIO25	UNALLOCATED	DO NOT CONNECT
Pin 23	GPIO11	UNALLOCATED	DO NOT CONNECT
Pin 24	GPIO8	UNALLOCATED	DO NOT CONNECT
Pin 25	GND	UNALLOCATED	DO NOT CONNECT
Pin 26	GPIO7	UNALLOCATED	DO NOT CONNECT
Pin 27	ID_SD	UNALLOCATED	DO NOT CONNECT
Pin 28	ID_SC	UNALLOCATED	DO NOT CONNECT
Pin 29	GPIO5	UNALLOCATED	DO NOT CONNECT
Pin 30	GND	UNALLOCATED	DO NOT CONNECT
Pin 31	GPIO6	UNALLOCATED	DO NOT CONNECT
Pin 32	GPIO12	UNALLOCATED	DO NOT CONNECT
Pin 33	GPIO13	UNALLOCATED	DO NOT CONNECT
Pin 34	GND	UNALLOCATED	DO NOT CONNECT
Pin 35	GPIO19	UNALLOCATED	DO NOT CONNECT
Pin 36	GPIO16	UNALLOCATED	DO NOT CONNECT
Pin 37	GPIO26	UNALLOCATED	DO NOT CONNECT
Pin 38	GPIO20	UNALLOCATED	DO NOT CONNECT
Pin 39	GND	UNALLOCATED	DO NOT CONNECT
Pin 40	GPIO21	UNALLOCATED	DO NOT CONNECT

Pin 1: NO EXTERNAL POWER. Unallocated positions: DO NOT CONNECT.

Source S7: official Pi 4 Model B datasheet, release 1.1, Fig. 3, p9.

BCM GPIO / board functions only; no new TARK allocation.

Power/ground wire IDs: S2. Purchased power method remains HOLD.



HOLD PORTS / NORMALIZED OBSERVATION BOUNDARIES		
Port	Electrical status	Software boundary / no hardware claim
MLX90640 VCC/VIN	OPEN - H09	Concrete reader and health contract; no fabricated temperatures.
BNO055 VCC/VIN	OPEN - H10	Concrete reader and health contract; physical calibration pending.
PS1 IN- / K1 coil return	Unresolved termination - S2	Do not complete unknown returns from an assumed common battery reference.
GNSS1 -> location service	USB serial cable only - H18/H19	Reader -> NMEA checksum -> normalized fix -> health -> vehicle location -> map.
LD2450 -> perception	RADAR-TX / RADAR-RX	Serial bytes -> bounded frame parser -> normalized reports -> health/freshness -> perception -> tracking -> TTC -> stopping requirement -> PV-SOE.
Pi to ESP32	No physical GPIO/cable allocation - H26	Configured serial transport -> Protocol V1 -> supervisor; see S5.

U2 J1 - ALL 22 POSITIONS IN PHYSICAL NUMBER ORDER		
J1	Controlled signal	Net / connection status
J1-1	UNALLOCATED	DO NOT CONNECT
J1-2	UNALLOCATED	DO NOT CONNECT
J1-3	UNALLOCATED	DO NOT CONNECT
J1-4	GPIO4	ENC-L-A
J1-5	GPIO5	ENC-L-B
J1-6	GPIO6	ENC-R-A
J1-7	GPIO7	ENC-R-B
J1-8	UNALLOCATED	DO NOT CONNECT
J1-9	UNALLOCATED	DO NOT CONNECT
J1-10	UNALLOCATED	DO NOT CONNECT
J1-11	UNALLOCATED	DO NOT CONNECT
J1-12	UNALLOCATED	DO NOT CONNECT
J1-13	UNALLOCATED	DO NOT CONNECT
J1-14	UNALLOCATED	DO NOT CONNECT
J1-15	GPIO9	ESP-PWM1
J1-16	GPIO10	ESP-DIR1
J1-17	GPIO11	ESP-PWM2
J1-18	GPIO12	ESP-DIR2
J1-19	GPIO13	ESTOP-AUX-STATUS
J1-20	UNALLOCATED	DO NOT CONNECT
J1-21	5V	N-LOGIC5+
J1-22	GND	N-LOGIC-GND

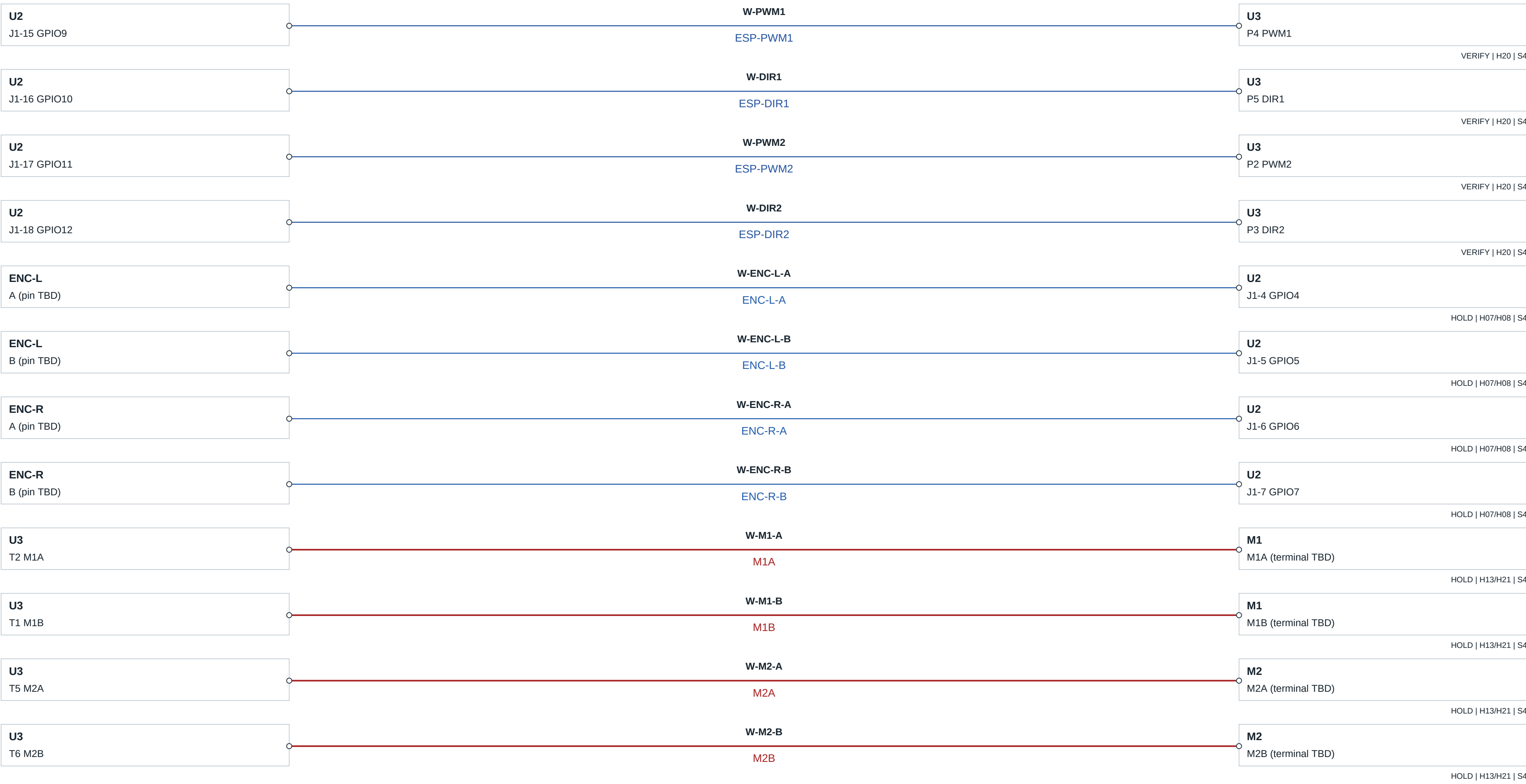
UNALLOCATED rows contain no invented GPIO identity.
Header viewing orientation: VERIFY purchased board.

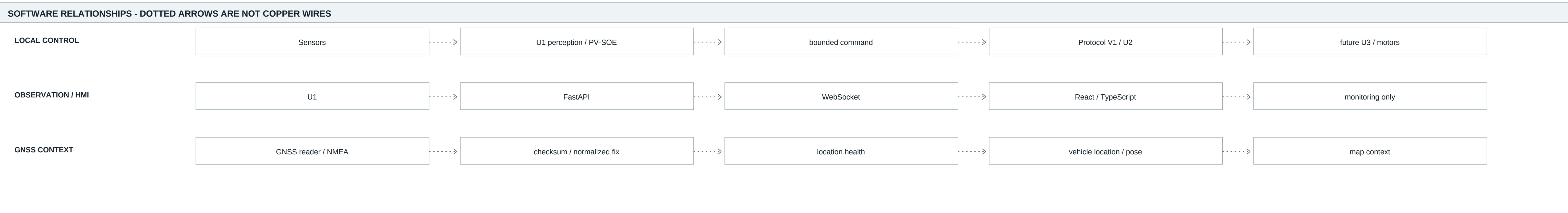
DO NOT CONNECT / HOLD
U2 J1 positions 1, 2, 3, 8, 9, 10, 11, 12, 13, 14 and 20: UNALLOCATED / DO NOT CONNECT.
The target connector has 22 positions; physical identification remains a commissioning task.
Encoder supply and signal compatibility remain HOLD.
No motion, firmware flash, GPIO output or physical measurement was performed for this package.
Wire lanes are authoritative pin-to-pin representations. Repeated component references do not create new components.

U3 MDD10A - LOGIC AND POWER INTERFACES		
Point	Signal	Net
P1	GND	N-LOGIC-GND
P2	PWM2	ESP-PWM2
P3	DIR2	ESP-DIR2
P4	PWM1	ESP-PWM1
P5	DIR1	ESP-DIR1
T1	M1B	M1B
T2	M1A	M1A
T3	POWER+	N-TRACTION12+
T4	POWER-	N-TRACTION-
T5	M2A	M2A
T6	M2B	M2B

T4 = N-TRACTION-; no copper bridge to logic ground.
T3 HOLD: physical E-stop isolation evidence required.

ENCODERS / MOTOR SEPARATION
M1/M2 motor current stays on M1A/M1B and M2A/M2B.
ENC-L A/B -> J1-4 GPIO4 / J1-5 GPIO5.
ENC-R A/B -> J1-6 GPIO6 / J1-7 GPIO7.
Encoder VCC: OPEN / HOLD H07/H08.
Encoder GND: HOLD / PHYSICAL PIN VERIFY. Logic reference only if purchased interface supports it. See W-GND-008/009 on S2.
No connector pin numbers, wire colors, polarity or pulse counts are assigned.
U2 J1-19 GPIO13 is diagnostic only. It is not the E-stop safety-current conductor.
All Phase 1 applied left/right motor outputs remain zero.
U2 power J1-21/J1-22 and U3 P1/T3/T4 are on S2.





CANONICAL ELECTRICAL NET REGISTER				
ID	One canonical net name	Resolved	HOLD	Detail
N001	N-BAT+	0	2	S2
N002	N-BMS+	0	2	S2
N003	N-PROTECTED12+	0	4	S2
N004	N-TRACTION12+	1	1	S2
N005	N-TRACTION-	1	2	S2
N006	N-LOGIC12+	0	2	S2
N007	N-LOGIC5+	3	1	S2
N008	N-LOGIC-GND	5	5	S2
N009	N-ESTOP-FUSED+	0	2	S2
N010	N-ESTOP-SWITCHED+	0	2	S2
N011	K1-COIL+	0	3	S2
N012	K1-COIL-	0	2	S2
N013	K1-AUX-RAW	0	2	S2
N014	ESTOP-AUX-STATUS	1	1	S2
N015	I2C-SDA	1	2	S3
N016	I2C-SCL	1	2	S3
N017	RADAR-TX	2	0	S3
N018	RADAR-RX	2	0	S3
N019	ESP-PWM1	2	0	S4
N020	ESP-DIR1	2	0	S4
N021	ESP-PWM2	2	0	S4
N022	ESP-DIR2	2	0	S4
N023	ENC-L-A	1	1	S4
N024	ENC-L-B	1	1	S4
N025	ENC-R-A	1	1	S4
N026	ENC-R-B	1	1	S4
N027	M1A	1	1	S4
N028	M1B	1	1	S4
N029	M2A	1	1	S4
N030	M2B	1	1	S4
N031	PS1-IN-RETURN-HOLD	0	1	S2

Cable assemblies: CAMERA-USB, GNSS-USB, GNSS-ANTENNA. These are interfaces, not single electrical nets.

PS1-IN-RETURN-HOLD is a singleton open input; no return bridge.

N-BAT- is not added as an alias of N-TRACTION-. K1-TRACTION+ is not added as an alias of N-TRACTION12+.

Resolved = controlled pin identifier only, NOT hardware verification. HOLD = functional / physical pin unresolved.

DO NOT CONNECT / HOLD REGISTER AND DRAWING LEGEND		
ID	Item / interface	Status
H01	U1 Pin 1	NO EXTERNAL POWER
H02	U2 J1 unallocated	DO NOT CONNECT
H03	B1 / BMS	TBD / VERIFY
H04	F1 / F2 / F3	TBD / VERIFY
H05	PS1	TBD / VERIFY
H06	U1 / U2 power	TBD / VERIFY
H07	ENC-L VCC / signals	TBD / VERIFY
H08	ENC-R VCC / signals	TBD / VERIFY
H09	MLX90640 VCC / I2C / GND	TBD / VERIFY
H10	BNO055 VCC / I2C / GND	TBD / VERIFY
H11	K1 coil	TBD / VERIFY
H12	CD1 and coil return	TBD / VERIFY
H13	K1 main contact / U3 T3	HOLD
H14	D1/S1 suppression	TBD / VERIFY
H15	K1 auxiliary / SC1	TBD / VERIFY
H16	LD2450 interface	TBD / VERIFY
H17	CAM1	TBD / VERIFY
H18	USB interfaces	TBD / VERIFY
H19	GNSS1 / ANT1	TBD / VERIFY
H20	U2 / U3	TBD / VERIFY
H21	M1 / M2	TBD / VERIFY
H22	Harnesses / connectors	TBD / VERIFY
H23	PCB footprints	TBD / VERIFY
H24	PCB mechanics	TBD / VERIFY
H25	PCB routing classes	TBD / VERIFY
H26	Pi to ESP32 transport	TBD / VERIFY
H27	U1 unallocated header	DO NOT CONNECT

Full hold requirements are in the workbook and HOLD_REGISTER.md.

JUNCTION DOT = ELECTRICAL CONNECTION ONLY.

BLACK DASHED = MECHANICAL ONLY. Crossings without dots are unconnected.

USB/RF cable lanes are dashed and explicitly labelled by cable ID.

Protocol V1: COBS, CRC-32C, canonical CBOR, sequence, timestamp, expiry, configuration hash, identity gate, ACK/NACK/STATUS/HEARTBEAT.

The dashboard cannot command motors, override PV-SOE or bypass E-stop. GNSS remains observational.

Electrical wire classes: HIGH CURRENT / TRACTION RETURN / LOGIC POWER / LOGIC RETURN / I2C / UART / ENCODER / DIGITAL SIGNAL / SAFETY CONTROL / DIAGNOSTIC